Introduction to Data Science

Mid-Term Project — Data Preprocessing & Exploratory Analysis

Dataset: Adult / Census Income (UCI Machine Learning Repository)

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Section: L | Date: 29/07/2026

1. About the Dataset

This project uses the Adult (also called Census Income) dataset from the UCI Machine Learning Repository. It contains 32,561 records and 15 columns describing demographic and employment attributes of individuals drawn from the 1994 U.S. Census, including age, workclass, education, marital status, occupation, race, sex, capital gain/loss, hours worked per week, and native country. The target variable is income, a two-level categorical label indicating whether a person earns more than $50,000 per year (>50K) or not (<=50K). The dataset is well suited to a preprocessing project. It ships with genuine missing values stored as "?", mixed data types, and a class imbalance, giving a realistic basis for cleaning, transformation, and exploratory analysis.

Source: Becker, B. & Kohavi, R. (1996). Adult [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5XW20 | https://archive.ics.uci.edu/dataset/2/adult

2. Loading the Original Dataset

R Code

```
adult_income <- read_csv("adult_income.csv")
dim(adult_income)
str(adult_income)
summary(adult_income)
```

Output

```
> dim(adult_income)
[1] 32561    15

> str(adult_income)
spc_tbl_ [32,561 × 15] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ age           : num [1:32561] 39 50 38 53 28 37 49 52 31 42 ...
 $ workclass     : chr [1:32561] "State-gov" "Self-emp-not-inc" "Private" "Private" ...
 $ fnlwgt        : num [1:32561] 77516 83311 215646 234721 338409 ...
 $ education     : chr [1:32561] "Bachelors" "Bachelors" "HS-grad" "11th" ...
 $ education_num : num [1:32561] 13 13 9 7 13 14 5 9 14 13 ...
 $ marital_status: chr [1:32561] "Never-married" "Married-civ-spouse" "Divorced" "Married-
civ-spouse" ...
 $ occupation    : chr [1:32561] "Adm-clerical" "Exec-managerial" "Handlers-cleaners"
"Handlers-cleaners" ...
 $ relationship  : chr [1:32561] "Not-in-family" "Husband" "Not-in-family" "Husband" ...
 $ race          : chr [1:32561] "White" "White" "White" "Black" ...
 $ sex           : chr [1:32561] "Male" "Male" "Male" "Male" ...
 $ capital_gain  : num [1:32561] 2174 0 0 0 0 ...
 $ capital_loss  : num [1:32561] 0 0 0 0 0 0 0 0 0 0 ...
 $ hours_per_week: num [1:32561] 40 13 40 40 40 40 16 45 50 40 ...
```

```
 $ native_country: chr [1:32561] "United-States" "United-States" "United-States" "United-
States" ...
 $ income        : chr [1:32561] "<=50K" "<=50K" "<=50K" "<=50K" ...
 - attr(*, "spec")=
  .. cols(
  ..   age = col_double(),
  ..   workclass = col_character(),
  ..   fnlwgt = col_double(),
  ..   education = col_character(),
  ..   education_num = col_double(),
  ..   marital_status = col_character(),
  ..   occupation = col_character(),
  ..   relationship = col_character(),
  ..   race = col_character(),
  ..   sex = col_character(),
  ..   capital_gain = col_double(),
  ..   capital_loss = col_double(),
  ..   hours_per_week = col_double(),
  ..   native_country = col_character(),
  ..   income = col_character()
  .. )
 - attr(*, "problems")=<pointer: 0x7fe62887f4c0>

> summary(adult_income)
      age            workclass         fnlwgt            education     education_num
 Min.   :17.00   Length   :32561   Min.   :  12285   Length   :32561   Min.   : 1.00
 1st Qu.:28.00   N.unique :    9   1st Qu.: 117827   N.unique :   16   1st Qu.: 9.00
 Median :37.00   N.blank  :    0   Median : 178356   N.blank  :    0   Median :10.00
 Mean   :38.58   Min.nchar:    1   Mean   : 189778   Min.nchar:    3   Mean   :10.08
 3rd Qu.:48.00   Max.nchar:   16   3rd Qu.: 237051   Max.nchar:   12   3rd Qu.:12.00
 Max.   :90.00                     Max.   :1484705                     Max.   :16.00
   marital_status      occupation       relationship          race              sex
 Length   :32561   Length   :32561   Length   :32561   Length   :32561   Length   :32561
 N.unique :    7   N.unique :   15   N.unique :    6   N.unique :    5   N.unique :    2
 N.blank  :    0   N.blank  :    0   N.blank  :    0   N.blank  :    0   N.blank  :    0
 Min.nchar:    7   Min.nchar:    1   Min.nchar:    4   Min.nchar:    5   Min.nchar:    4
 Max.nchar:   21   Max.nchar:   17   Max.nchar:   14   Max.nchar:   18   Max.nchar:    6

  capital_gain    capital_loss    hours_per_week    native_country        income
 Min.   :    0   Min.   :   0.0   Min.   : 1.00   Length   :32561   Length   :32561
 1st Qu.:    0   1st Qu.:   0.0   1st Qu.:40.00   N.unique :   42   N.unique :    2
 Median :    0   Median :   0.0   Median :40.00   N.blank  :    0   N.blank  :    0
 Mean   : 1078   Mean   :  87.3   Mean   :40.44   Min.nchar:    1   Min.nchar:    4
 3rd Qu.:    0   3rd Qu.:   0.0   3rd Qu.:45.00   Max.nchar:   26   Max.nchar:    5
 Max.   :99999   Max.   :4356.0   Max.   :99.00
```

Interpretation: The dataset loads with 32,561 records and 15 variables, and read_csv correctly infers six numeric and nine character columns. Two details matter later: workclass reports nine unique values although only eight employment categories exist, the ninth being the literal "?" that this dataset uses to encode missing data, and capital_gain shows a maximum of 99,999 against a median of 0, indicating extreme right-skew. We used read_csv rather than base read.csv because it trims surrounding whitespace by default and does not silently convert text to factors.

3. Introducing Data Quality Issues

R Code

```
adult <- adult_income

sum(adult$workclass == "?")
sum(adult$occupation == "?")
sum(adult$native_country == "?")

adult$age[c(5, 20, 100)] <- NA
adult$hours_per_week[c(50, 120)] <- NA
adult$age[10] <- -8
adult$age[30] <- 150
adult$hours_per_week[60] <- -5
adult$sex[15] <- "Unknown"
adult$workclass[25] <- "private"
adult$workclass[40] <- " PRIVATE "
adult <- rbind(adult, adult[1:5, ])

adult$workclass[adult$workclass == "?"] <- NA
adult$occupation[adult$occupation == "?"] <- NA
adult$native_country[adult$native_country == "?"] <- NA

dim(adult)
sum(duplicated(adult))
colSums(is.na(adult))
```

Output

```
> sum(adult$workclass == "?")
[1] 1836
> sum(adult$occupation == "?")
[1] 1843
> sum(adult$native_country == "?")
[1] 583
> dim(adult) # 32566 rows now (32561 + 5 duplicates)
[1] 32566    15
> sum(duplicated(adult)) # 29 (24 already in the data + 5 we added)
[1] 29

> colSums(is.na(adult)) # NA count per column after conversion
           age      workclass         fnlwgt      education  education_num marital_status
             4           1836              0              0              0              0
    occupation   relationship           race            sex   capital_gain   capital_loss
          1843              0              0              0              0              0
hours_per_week native_country         income
             2            583              0
```

Interpretation: The counts confirm the published dataset already stores 1,836 missing workclass, 1,843 missing occupation and 583 missing native_country values as the text "?". Converting these to NA is necessary because R's missing-value functions recognise only NA, so leaving them as text would report zero missing data while treating "?" as a legitimate ninth category. On top of these we injected missing values, impossible values, an invalid category, inconsistent text and five duplicate rows. The row count rises to 32,566 and duplicated reports 29 rows, the 24 already present in the published data plus the five we added.

4. Missing Value Inspection

R Code

```
missing_values <- colSums(is.na(adult))
print(missing_values)
barplot(missing_values,
        las  = 2,
        col  = "steelblue",
        main = "Missing Values by Column",
        ylab = "Count")

# Is the missingness random?
sum(is.na(adult$workclass) & is.na(adult$occupation))

mean(adult$age[is.na(adult$workclass)] < 25 |
     adult$age[is.na(adult$workclass)] > 65, na.rm = TRUE) * 100
mean(adult$age[!is.na(adult$workclass)] < 25 |
     adult$age[!is.na(adult$workclass)] > 65, na.rm = TRUE) * 100
```

Output

```
> print(missing_values)
           age      workclass         fnlwgt      education  education_num marital_status
             4           1836              0              0              0              0
    occupation   relationship           race            sex   capital_gain   capital_loss
          1843              0              0              0              0              0
hours_per_week native_country         income
             2            583              0
> sum(is.na(adult$workclass) & is.na(adult$occupation))
[1] 1836
> mean(adult$age[is.na(adult$workclass)] < 25 |
+        adult$age[is.na(adult$workclass)] > 65, na.rm = TRUE) * 100
[1] 51.08932
> mean(adult$age[!is.na(adult$workclass)] < 25 |
+        adult$age[!is.na(adult$workclass)] > 65, na.rm = TRUE) * 100
[1] 18.85048
```

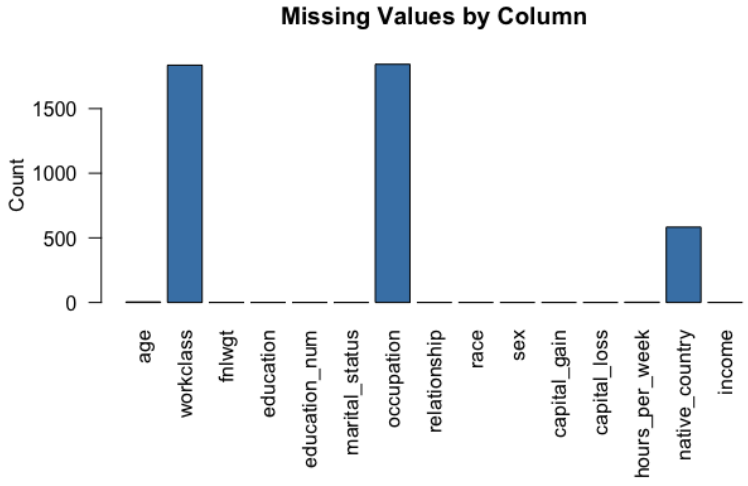


Fig. 1. Missing Values by Column

Interpretation: The chart shows missingness confined to three columns, with workclass and occupation at roughly 5.6% each and native_country at 1.8%. The pattern is not random. The two employment columns are missing together in all 1,836 cases, and 51.09% of those records are aged under 25 or over 65 against only 18.85% of the rest. These are students and retirees, so the value is missing because the person has no job. The missingness therefore carries information, which rules out deleting these rows, as that would systematically remove the youngest and oldest respondents.

5. Fixing Inconsistent Text Format

R Code

```
adult$workclass <- trimws(adult$workclass)
adult$workclass[tolower(adult$workclass) == "private"] <- "Private"
adult$sex <- trimws(adult$sex)

table(adult$workclass)
table(adult$sex)
```

Output

```
> table(adult$workclass)

     Federal-gov        Local-gov     Never-worked          Private     Self-emp-inc
             960             2093                7            22700             1116
Self-emp-not-inc        State-gov      Without-pay
            2541             1299               14

> table(adult$sex)

 Female    Male Unknown
  10772   21793       1
```

Interpretation: Trimming whitespace and mapping the case variants of "private" back to "Private" restores workclass to its eight documented categories. We used a targeted correction rather than a blanket case transformation, which would have rewritten labels that were already correct and turned "Self-emp-not-inc" into inconsistent forms. Whitespace is trimmed before any comparison because " PRIVATE " and "PRIVATE" are different strings to R. The invalid "Unknown" value in sex remains at this stage, since it is not a formatting problem but a value outside the variable's domain.

6. Detecting and Fixing Invalid Data

R Code

```
adult[adult$age < 0, ]
adult[adult$age > 100, ]
adult[adult$hours_per_week < 0, ]
summary(adult$age)

adult$age[adult$age < 0] <- NA
adult$age[adult$age > 100] <- NA
adult$hours_per_week[adult$hours_per_week < 0] <- NA
adult$sex[adult$sex == "Unknown"] <- NA

sum(is.na(adult$age))              # 4 -> 6
sum(is.na(adult$hours_per_week))   # 2 -> 3
```

Output

```
> adult[adult$age < 0, ]
# A tibble: 5 × 15
    age workclass fnlwgt education education_num marital_status occupation relationship
race  sex
  <dbl> <chr>      <dbl> <chr>             <dbl> <chr>          <chr>      <chr>
<chr> <chr>
1    NA NA            NA NA                   NA NA             NA         NA           NA
NA
2    -8 Private   159449 Bachelors            13 Married-civ-s… Exec-mana… Husband
White Male
3    NA NA            NA NA                   NA NA             NA         NA           NA
NA
4    NA NA            NA NA                   NA NA             NA         NA           NA
NA
5    NA NA            NA NA                   NA NA             NA         NA           NA
NA
# i 5 more variables: capital_gain <dbl>, capital_loss <dbl>, hours_per_week <dbl>,
#   native_country <chr>, income <chr>
> adult[adult$age > 100, ]
# A tibble: 5 × 15
    age workclass fnlwgt education education_num marital_status occupation relationship
race  sex
  <dbl> <chr>      <dbl> <chr>             <dbl> <chr>          <chr>      <chr>
<chr> <chr>
1    NA NA            NA NA                   NA NA             NA         NA           NA
NA
2    NA NA            NA NA                   NA NA             NA         NA           NA
NA
3   150 Private   193366 HS-grad              9 Married-civ-s… Craft-rep… Husband
White Male
4    NA NA            NA NA                   NA NA             NA         NA           NA
NA
5    NA NA            NA NA                   NA NA             NA         NA           NA
NA
# i 5 more variables: capital_gain <dbl>, capital_loss <dbl>, hours_per_week <dbl>,
#   native_country <chr>, income <chr>
> adult[adult$hours_per_week < 0, ]
# A tibble: 3 × 15
    age workclass fnlwgt education education_num marital_status occupation relationship
race  sex
  <dbl> <chr>      <dbl> <chr>             <dbl> <chr>          <chr>      <chr>
<chr> <chr>
1    NA NA            NA NA                   NA NA             NA         NA           NA
NA
2    30 Private   188146 HS-grad              9 Married-civ-s… Machine-o… Husband
White Male
3    NA NA            NA NA                   NA NA             NA         NA           NA
NA
# i 5 more variables: capital_gain <dbl>, capital_loss <dbl>, hours_per_week <dbl>,
#   native_country <chr>, income <chr>
> summary(adult$age)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
  -8.00   28.00   37.00   38.58   48.00  150.00       4

> sum(is.na(adult$age))              # 4 -> 6
[1] 6
> sum(is.na(adult$hours_per_week))   # 2 -> 3
[1] 3
```

Interpretation: The summary exposes the impossible values: a minimum age of −8 and a maximum of 150 against a plausible range of 17 to 90, plus a value of −5 hours worked. The NA rows in the subset output are an artefact of comparing a column that already contains NA, not additional invalid records; the genuine cases are the single rows showing −8, 150 and −5. We treat these as invalid rather than as outliers, since an outlier is a real but unusual observation while an invalid entry is a recording error. Each offending cell is set to NA rather than deleting the row, preserving the fourteen other valid fields, and missing values rise from 4 to 6 in age and 2 to 3 in hours_per_week.

7. Handling Missing Values (Imputation)

R Code

```
adult$age[is.na(adult$age)] <-
  median(adult$age, na.rm = TRUE)
adult$hours_per_week[is.na(adult$hours_per_week)] <-
  median(adult$hours_per_week, na.rm = TRUE)

Mode <- function(x) {
  ux <- na.omit(unique(x))
  ux[which.max(tabulate(match(x, ux)))]
}

adult$workclass[is.na(adult$workclass)] <- Mode(adult$workclass)
adult$occupation[is.na(adult$occupation)] <- Mode(adult$occupation)
adult$native_country[is.na(adult$native_country)] <- Mode(adult$native_country)
adult$sex[is.na(adult$sex)] <- Mode(adult$sex)
sum(is.na(adult))
```

Output

```
> sum(is.na(adult))
[1] 0
```

Interpretation: All missing values are resolved and sum(is.na(adult)) returns zero. The method was chosen per column according to data type and distribution. For age and hours_per_week we used the median rather than the mean because both distributions are right-skewed and the mean would be pulled upward by the tail; for hours_per_week the median of 40 is also the most frequent value, making it the most probable single guess. For the categorical columns neither mean nor median is defined, so we used the mode through a small custom function, as base R provides none. We rejected listwise deletion because it would discard roughly 7.4% of the data and, since the missingness concentrates among the youngest and oldest, would bias every subsequent statistic. One limitation: assigning all 1,836 missing employment records to Private attributes a job to people who likely had none, and an explicit "Unknown" level would have preserved that information.

8. Removing Duplicate Rows

R Code

```
before <- nrow(adult)
adult  <- adult[!duplicated(adult), ]
after  <- nrow(adult)

cat("Rows before:", before, "\n")
cat("Rows after:",  after,  "\n")
```

Output

```
> cat("Rows before:", before, "\n")
Rows before: 32566
> cat("Rows after:",  after,  "\n")
Rows after: 32537
```

Interpretation: Removing exact duplicates reduces the dataset from 32,566 to 32,537 rows, eliminating 29 records: the 24 already present in the published data plus the five we added. Duplicates inflate the apparent frequency of whatever pattern they contain and violate the independence assumption behind most statistical tests. The !duplicated approach keeps the first occurrence of each identical set, which is appropriate because the rows match in every field. Deduplication is performed after text standardisation, because rows differing only in capitalisation are not recognised as duplicates until the text is unified.

9. Outlier Detection

R Code

```
boxplot(adult$age, main = "Age Outliers", col = "lightgreen")
boxplot(adult$capital_gain, main = "Capital Gain Outliers", col = "orange")

cap <- quantile(adult$capital_gain, 0.99)
cap
sum(adult$capital_gain > cap)
sum(adult$capital_gain == 99999)
round(mean(adult$capital_gain == 0) * 100, 1)

adult$capital_gain[adult$capital_gain > cap] <- cap
```

Output

```
 > cap
   99%
 15024
 > sum(adult$capital_gain > cap)
 [1] 261
 > sum(adult$capital_gain == 99999)
 [1] 159
 > round(mean(adult$capital_gain == 0) * 100, 1)
 [1] 91.7
```

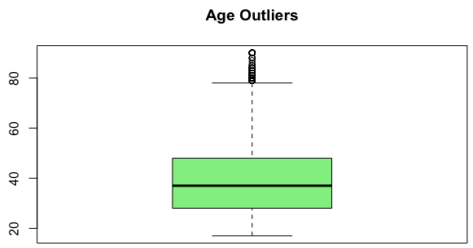


Fig. 2. Age Distribution

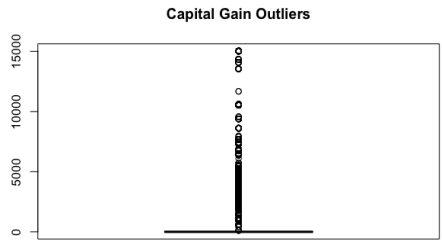


Fig. 3. Capital Gain Distribution

Interpretation: Age is broadly symmetric with a few high values that the 1.5 IQR rule flags but which are entirely plausible ages, so we left the column unchanged. capital_gain is a different case: 91.7% of its values are zero and the remainder stretches to 99,999, so we capped it at its 99th percentile of 15,024, affecting 261 records. Capping was preferred over deletion because the affected rows hold valid data in every other field. Of the 261 capped records, 159 sat at exactly 99,999, which is a census top-code rather than a real amount, so the upper tail of this variable is censored rather than measured.

10. Converting Between Numeric and Categorical (and Normalisation)

R Code

```
# Categorical text -> factors
adult$income    <- as.factor(adult$income)
adult$sex       <- as.factor(adult$sex)
adult$workclass <- as.factor(adult$workclass)

# Binning: continuous age -> ordered age bands
adult$Age_Group <- cut(adult$age,
                       breaks = c(0, 25, 45, 65, 100),
                       labels = c("Young", "Adult", "Middle-aged", "Senior"),
                       include.lowest = TRUE)

# Label encoding: binary category -> 0/1
adult$Sex_Code <- ifelse(adult$sex == "Male", 1, 0)

# Normalisation: min-max scale hours_per_week onto [0, 1]
adult$Hours_Normalized <-
  (adult$hours_per_week - min(adult$hours_per_week)) /
  (max(adult$hours_per_week) - min(adult$hours_per_week))

table(adult$Age_Group) # counts per age band
summary(adult$Hours_Normalized) # confirms the [0, 1] range
```

Output

```
> table(adult$Age_Group) # counts per age band
      Young       Adult Middle-aged      Senior
       6400       16516        8464        1157
> summary(adult$Hours_Normalized) # confirms the [0, 1] range
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.0000  0.3980  0.3980  0.4025  0.4490  1.0000
```

Interpretation: Converting income, sex and workclass to factors tells R they are categorical with fixed levels, which is what allows table and the grouped summaries to work correctly. Binning age into four ordered bands produces 6,400 Young, 16,516 Adult, 8,464 Middle-aged and 1,157 Senior records, trading numerical precision for interpretability. Label encoding maps Male to 1 and Female to 0; this is safe only because sex has two levels, since coding a multi-level nominal variable as 1 through 5 would falsely imply an order and equal spacing. Min-max normalisation rescales hours_per_week to exactly 0 and 1, and we chose it over z-score standardisation because we wanted a bounded range. The identical 0.3980 at both the first quartile and the median is not an error: 40 hours dominates the column, and (40 − 1) divided by (99 − 1) equals 0.398.

11. Checking and Fixing Class Imbalance

R Code

```
table(adult$income)
barplot(table(adult$income),
        col  = c("skyblue", "tomato"),
        main = "Income Distribution")

set.seed(123) # reproducible sampling
low  <- adult[adult$income == "<=50K", ]
high <- adult[adult$income == ">50K",  ]
n    <- min(nrow(low), nrow(high)) # size of the smaller class

balanced <- rbind(low[sample(nrow(low), n), ], high)
table(balanced$income) # now 7839 vs 7839
```

Output

```
> table(adult$income)
<=50K  >50K
24698  7839

> table(balanced$income)
<=50K  >50K
 7839  7839
```

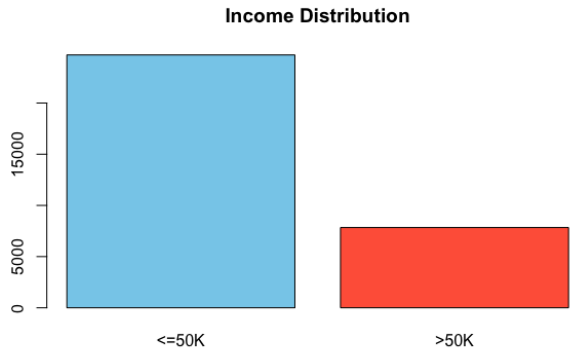


Fig. 4. Income Class Distribution Before Balancing

Interpretation: The target is imbalanced at 24,698 records against 7,839, roughly 76 to 24. This matters because a model could reach 76% accuracy by predicting the majority class for every record while identifying no high earners at all. Random undersampling draws 7,839 majority records to match the minority exactly, giving a balanced 7,839 to 7,839 set. Undersampling introduces no synthetic data but discards 16,859 genuine records; oversampling would retain them at the risk of overfitting duplicated rows. Balancing is a modelling step rather than a cleaning step, so all following analysis uses the full dataset to keep the statistics representative.

12. Descriptive Statistics Across Two Income Classes

R Code

```
adult %>%
  group_by(income) %>%
  summarise(
    Average_Age   = mean(age),
    Median_Age    = median(age),
    SD_Age        = sd(age),
    Average_Hours = mean(hours_per_week),
    Education     = mean(education_num)
)
```

Output

```
> adult %>%
+   group_by(income) %>%
+   summarise(
+     Average_Age   = mean(age),
+     Median_Age    = median(age),
+     SD_Age        = sd(age),
+     Average_Hours = mean(hours_per_week),
+     Education     = mean(education_num)
+ )
# A tibble: 2 × 6
  income Average_Age Median_Age SD_Age Average_Hours Education
  <fct>        <dbl>      <dbl>  <dbl>         <dbl>     <dbl>
1 <=50K         36.8         34   14.0          38.8      9.60
2 >50K          44.2         44   10.5          45.5     11.6
```

Interpretation: Higher earners are on average 44.2 years old against 36.8, work 45.5 hours per week against 38.8, and hold 11.6 years of education on the ordinal scale against 9.6. The median ages of 44 and 34 confirm the gap is not driven by a few extreme values. The age standard deviation is also lower in the higher group at 10.5 against 14.0, meaning high earners cluster into a narrower band while lower earners span the full working-age range.

13. Comparing Mean Hours per Week Between Income Groups

R Code

```
t.test(hours_per_week ~ income, data = adult)

boxplot(hours_per_week ~ income,
        data = adult,
        col  = c("lightblue", "pink"),
        main = "Working Hours by Income")
```

Output

```
> t.test(hours_per_week ~ income, data = adult)

        Welch Two Sample t-test

data:  hours_per_week by income
t = -45.096, df = 14568, p-value < 2.2e-16
alternative hypothesis: true difference in means between group <=50K and group >50K is not
equal to 0
95 percent confidence interval:
 -6.918909 -6.342496
sample estimates:
mean in group <=50K  mean in group >50K
            38.8427              45.4734
```

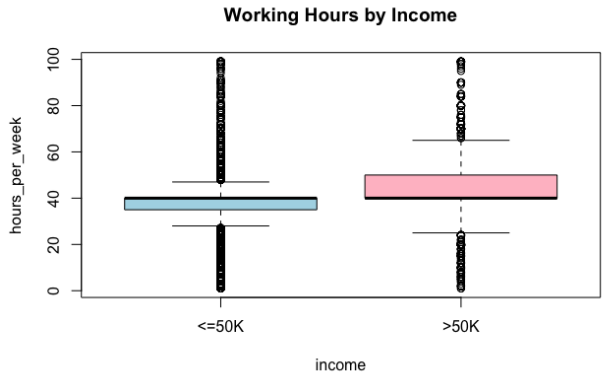


Fig. 5. Working Hours per Week by Income Group

Interpretation: The two means differ by 6.63 hours per week, and the Welch t-test confirms the difference is significant with t = −45.096 and a p-value below 2.2e-16. The 95% confidence interval of −6.92 to −6.34 excludes zero, so the difference is real and close to six and a half hours. We used Welch rather than Student's t-test because it does not assume equal variances, which the group standard deviations show do not hold here. The boxplot adds a nuance the means conceal: both medians are 40 hours, so the difference lies in the shape of the distribution rather than in the typical respondent.

14. Comparing the Spread of Age Across Work Classes

R Code

```
adult %>%
  group_by(workclass) %>%
  summarise(
    Mean_Age = mean(age),
    SD       = sd(age)
  )

boxplot(age ~ workclass,
        data = adult,
        las  = 2,
        col  = "lightgreen")
```

Output

```
> adult %>%
+   group_by(workclass) %>%
+   summarise(
+     Mean_Age = mean(age),
+     SD       = sd(age)
+   )
# A tibble: 8 × 3
  workclass        Mean_Age    SD
  <fct>               <dbl> <dbl>
1 Federal-gov          42.6 11.5
2 Local-gov            41.8 12.3
3 Never-worked         20.6  4.61
4 Private              37.1 13.6
5 Self-emp-inc         46.0 12.6
6 Self-emp-not-inc     45.0 13.3
7 State-gov            39.4 12.4
8 Without-pay          47.8 21.1
```

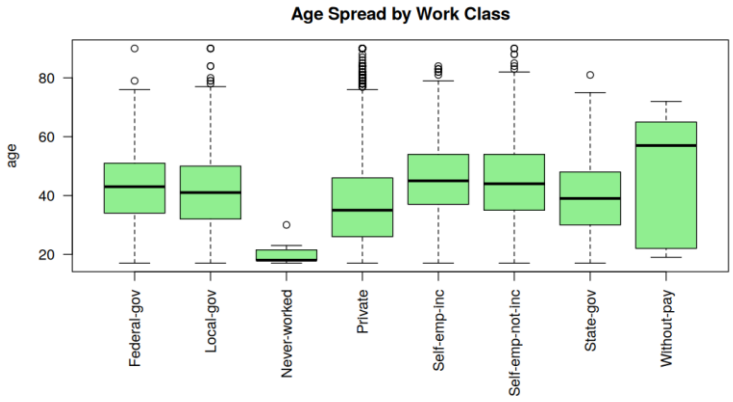


Fig. 6. Age Spread Across Work Classes

Interpretation: Mean age and spread both vary by employment category. The self-employed are the oldest at 46.0 and 45.0 years, while private sector workers are the youngest of the large categories at 37.1. Federal-gov is the most tightly clustered with a standard deviation of 11.5, consistent with structured entry requirements and defined retirement policies, whereas private and self-employed work admit people at any stage of life. Never-worked and Without-pay hold only 7 and 14 records, so their figures are too unstable to interpret. The Private spread is also partly inflated by the mode imputation, which added 1,836 records skewed toward the very young and old and raised its standard deviation from 12.83 to 13.57.

15. Filtering Data

R Code

```
# filter on two conditions at once
high_earners <- adult %>% filter(income == ">50K", age > 40)
nrow(high_earners)

overworked <- adult %>% filter(hours_per_week > 60, workclass == "Private")
nrow(overworked)

# filter then aggregate
adult %>% filter(Age_Group == "Senior") %>% count(income)

# filter, sort and select specific columns
adult %>%
  filter(income == ">50K") %>%
  arrange(desc(hours_per_week)) %>%
  select(age, workclass, education_num, hours_per_week, income) %>%
  head(5)
```

Output

```
> nrow(high_earners)
[1] 4750
> nrow(overworked)
[1] 609

> adult %>% filter(Age_Group == "Senior") %>% count(income)
# A tibble: 2 × 2
  income     n
  <fct>  <int>
1 <=50K    924
2 >50K     233

> adult %>%
+   filter(income == ">50K") %>%
+   arrange(desc(hours_per_week)) %>%
+   select(age, workclass, education_num, hours_per_week, income) %>%
+   head(5)
# A tibble: 5 × 5
    age workclass        education_num hours_per_week income
  <dbl> <fct>                    <dbl>          <dbl> <fct>
1    37 Private                      9             99 >50K
2    25 Private                     14             99 >50K
3    55 Self-emp-not-inc            10             99 >50K
4    37 Self-emp-inc                16             99 >50K
5    43 Self-emp-inc                15             99 >50K
```

Interpretation: Filtering isolates the subsets relevant to a specific question rather than analysing all 32,537 records at once. Two conditions combined identify 4,750 respondents who are both high earners and over 40, and 609 private sector workers exceeding 60 hours per week. Chaining filter with count shows 233 of 1,157 Senior respondents earn above 50,000 dollars, a rate of 20% against the 24% population rate. The final chain filters, sorts and selects columns to surface the highest-hours high earners, where 99 is the upper bound of the variable as recorded. We used dplyr's filter rather than bracket subsetting because it drops NA rows instead of returning them as blanks.

16. Train / Test Split

R Code

```
set.seed(123)
index <- createDataPartition(adult$income, p = 0.80, list = FALSE)
train <- adult[index, ]
test  <- adult[-index, ]

dim(train) # ~80% of rows
dim(test) # ~20% of rows
```

Output

```
> dim(train)   # ~80% of rows
[1] 26031    18
> dim(test)    # ~20% of rows
[1] 6506   18
```

Interpretation: The data splits into 26,031 training and 6,506 testing records, an 80 to 20 partition. We used createDataPartition rather than a plain random sample because it stratifies, drawing 80% from within each income class separately so both sets retain the same class balance as the full data. This matters most for the minority >50K group, which an unstratified split could under-represent and make evaluation on that class unreliable. The seed fixes the sampling so the same split reproduces on every run.

17. Final Cleaned Dataset

R Code

```
summary(adult)
head(adult)
tail(adult)
```

Output

```
> summary(adult)
      age                 workclass         fnlwgt            education     education_num
 Min.   :17.00   Private         :24510   Min.   :  12285   Length   :32537   Min.   : 1.00
 1st Qu.:28.00   Self-emp-not-inc: 2539   1st Qu.: 117827   N.unique :   16   1st Qu.: 9.00
 Median :37.00   Local-gov       : 2093   Median : 178356   N.blank  :    0   Median :10.00
 Mean   :38.59   State-gov       : 1298   Mean   : 189781   Min.nchar:    3   Mean   :10.08
 3rd Qu.:48.00   Self-emp-inc    : 1116   3rd Qu.: 236993   Max.nchar:   12   3rd Qu.:12.00
 Max.   :90.00   Federal-gov     :  960   Max.   :1484705                     Max.   :16.00
                 (Other)         :   21
   marital_status      occupation       relationship          race           sex
 Length   :32537   Length   :32537   Length   :32537   Length   :32537   Female:10762
 N.unique :    7   N.unique :   14   N.unique :    6   N.unique :    5   Male  :21775
 N.blank  :    0   N.blank  :    0   N.blank  :    0   N.blank  :    0
 Min.nchar:    7   Min.nchar:    5   Min.nchar:    4   Min.nchar:    5
 Max.nchar:   21   Max.nchar:   17   Max.nchar:   14   Max.nchar:   18
```

```
  capital_gain      capital_loss     hours_per_week    native_country    income
 Min.   :    0.0   Min.   :   0.00   Min.   : 1.00   Length   :32537   <=50K:24698
 1st Qu.:    0.0   1st Qu.:   0.00   1st Qu.:40.00   N.unique :   41   >50K : 7839
 Median :    0.0   Median :   0.00   Median :40.00   N.blank  :    0
 Mean   :  634.3   Mean   :  87.37   Mean   :40.44   Min.nchar:    4
 3rd Qu.:    0.0   3rd Qu.:   0.00   3rd Qu.:45.00   Max.nchar:   26
 Max.   :15024.0   Max.   :4356.00   Max.   :99.00

       Age_Group        Sex_Code      Hours_Normalized
 Young      : 6400   Min.   :0.0000   Min.   :0.0000
 Adult      :16516   1st Qu.:0.0000   1st Qu.:0.3980
 Middle-aged: 8464   Median :1.0000   Median :0.3980
 Senior     : 1157   Mean   :0.6692   Mean   :0.4025
                     3rd Qu.:1.0000   3rd Qu.:0.4490
                     Max.   :1.0000   Max.   :1.0000

> head(adult)
# A tibble: 6 × 18
    age workclass fnlwgt education education_num marital_status occupation relationship
race  sex
  <dbl> <fct>      <dbl> <chr>             <dbl> <chr>          <chr>      <chr>
<chr> <fct>
1    39 State-gov  77516 Bachelors            13 Never-married  Adm-cleri… Not-in-fami…
White Male
2    50 Self-emp…  83311 Bachelors            13 Married-civ-s… Exec-mana… Husband
White Male
3    38 Private   215646 HS-grad               9 Divorced       Handlers-… Not-in-fami…
White Male
4    53 Private   234721 11th                  7 Married-civ-s… Handlers-… Husband
Black Male
5    37 Private   338409 Bachelors            13 Married-civ-s… Prof-spec… Wife
Black Fema…
6    37 Private   284582 Masters              14 Married-civ-s… Exec-mana… Wife
White Fema…
# ℹ 8 more variables: capital_gain <dbl>, capital_loss <dbl>, hours_per_week <dbl>,
#   native_country <chr>, income <fct>, Age_Group <fct>, Sex_Code <dbl>, Hours_Normalized
<dbl>

> tail(adult)
# A tibble: 6 × 18
    age workclass fnlwgt education education_num marital_status occupation relationship
race  sex
  <dbl> <fct>      <dbl> <chr>             <dbl> <chr>          <chr>      <chr>
<chr> <fct>
1    22 Private   310152 Some-col…           10 Never-married  Protectiv… Not-in-fami…
White Male
2    27 Private   257302 Assoc-ac…           12 Married-civ-s… Tech-supp… Wife
White Fema…
3    40 Private   154374 HS-grad               9 Married-civ-s… Machine-o… Husband
White Male
4    58 Private   151910 HS-grad               9 Widowed        Adm-cleri… Unmarried
White Fema…
5    22 Private   201490 HS-grad               9 Never-married  Adm-cleri… Own-child
White Male
6    52 Self-emp… 287927 HS-grad               9 Married-civ-s… Exec-mana… Wife
White Fema…
# ℹ 8 more variables: capital_gain <dbl>, capital_loss <dbl>, hours_per_week <dbl>,
#   native_country <chr>, income <fct>, Age_Group <fct>, Sex_Code <dbl>, Hours_Normalized
<dbl>
```

Interpretation: The final dataset holds 32,537 records and 18 columns, three more than the original because of the engineered features. No column reports missing values, age ranges from 17 to 90 with all impossible entries resolved, and capital_gain now caps at 15,024 rather than 99,999. workclass, sex and income are summarised by factor level rather than string length, confirming the type conversions took effect, with Private at 24,510 after imputation. Age_Group spans four bands, Hours_Normalized runs exactly 0 to 1, and Sex_Code has a mean of 0.6692, which is simply the male proportion and matches the counts of 21,775 against 10,762.

18. Summary and Conclusion

This project took the Adult Census Income dataset through a complete preprocessing and exploratory pipeline. Starting from 32,561 records, we injected missing values, impossible entries, an invalid category, inconsistent text and duplicate rows alongside the missing values the dataset already carried. We then repaired each in order: converting the "?" markers to NA, standardising text, quarantining impossible values as missing, imputing numeric gaps with the median and categorical gaps with the mode, and removing duplicates only after the text had been unified. We capped an extreme right-skewed variable, converted between numeric and categorical forms, normalised a continuous attribute, balanced the target classes and produced a stratified split, finishing at 32,537 complete records across 18 columns.

The analysis showed that higher earners are older, work about six and a half more hours per week and hold roughly two more years of education, with the difference in working hours confirmed significant by a Welch t-test. Age spread also varies markedly across employment sectors. The most instructive finding was methodological: missing employment data was concentrated among respondents under 25 and over 65, demonstrating that missingness can itself carry information rather than being merely an absence of it.

Two limitations should be noted. Mode imputation assigned 1,836 records to Private, attributing a job to respondents who likely had none, and capping capital_gain normalised 159 top-coded values, leaving that upper tail censored rather than measured. With these qualifications noted, the cleaned, balanced and partitioned dataset provides a sound basis for the classification modelling that would follow.